Assignment 1

1. Explain the Object-Oriented Programming paradigm. How object-oriented programming is different from structured programming.
2. What are the major features of OOP? Explain in detail with example .
3. What is a class and an object in Object-Oriented Programming? Explain class specification, access specifiers, member functions, inline functions, and data hiding with a suitable example.
4. Explain static data members and static member functions. Write a C++ program to count the number of objects created using a static data member.
5. What is a constructor? Explain various form of constructor with suitable example.
6. What is a destructor? Why is it needed? Explain when a destructor is invoked with a suitable C++ program.
7. Write short note on new and delete operator in c++.
8. What is function overloading and operator overloading? Explain difference between them.
9. Explain unary and binary operator overloading in C++. Write a programs to overload the unary - and binary + operators for a user-defined class.
10. What is a friend function? Explain how operator overloading can be implemented using friend functions. Write a program to overload the << and >> operators for input and output of objects using a friend function.
11. Explain the overloading of ++ and -- operators. Discuss the difference between prefix and postfix forms and their limitations.
12. Write a program to overload +, == , [] and = operator to concatenate, compare and copy two strings.
13. Explain this pointer in detail with suitable example.
14. What is data conversion between objects in C++? Explain class-to-class conversion using constructors and conversion functions with suitable examples. Also discuss memory leaks, their causes, and methods for tracing and preventing them.